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Physics-Informed Priors Improve Gravitational-Wave Constraints on the Nuclear Equation of State

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ABSTRACT

Gravitational-wave astronomy shows great promise in determining nuclear physics in a regime not accessible to terrestrial experiments. We introduce physics-informed priors constrained by nuclear theory and perturbative Quantum Chromodynamics calculations, as well as astrophysical measurements of neutron-star masses and radii. When these priors are used in gravitational-wave astrophysical inference, we show a significant improvement on nuclear equation of state constraints. Applying these to the first observed gravitational-wave binary neutron-star merger GW170817, the constraints on the radius of a $1.4\,M_{\odot}$ neutron star improve from $R_{1.4} = 12.54^{+1.05}_{-1.54}\,\mathrm{km}$ to $R_{1.4} = 12.11^{+0.91}_{-1.11}\,\mathrm{km}$ and those on the tidal deformability from $\tilde{\Lambda}_{1.186} < 720$ to $\tilde{\Lambda}_{1.186} = 384^{+306}_{-158}$ (90% confidence intervals) at the events measured chirp mass $\mathcal{M} = 1.186\,M_{\odot}$. We also show these priors can be used to perform model selection between binary neutron star and neutron star-black hole mergers; in the case of GW190425, the results provide only marginal evidence with a Bayes factor $\mathcal{BF} = 1.33$ in favour of the binary neutron star merger hypothesis. Given their ability to improve the astrophysical inference of binary mergers involving neutron stars, we advocate for these physics-informed priors to be used as standard in the literature and provide open-source code for reproducibility and adaptation of the method.

Keywords: neutron stars — gravitational waves — equation of state

1. INTRODUCTION

One of the main goals of gravitational wave (GW) astronomy is to infer constraints on the neutron-star equation of state (EOS) from observations of binary neutron star (BNS) mergers. The baryon density in neutron stars and their mergers can reach several times the nuclear saturation density $n_s = 0.16\,\mathrm{fm}^{-3}$ and temperatures of the order of tens of MeV (see, e.g., Baiotti & Rezzolla 2017; Paschalidis 2017; Radice et al. 2020, for reviews). Such densities and temperatures are inaccessible to terrestrial experiments and also to *ab initio* calculations due to the infamous fermionic sign problem, which prevents direct Quantum Chromodynamics (QCD) solutions on the lattice.

A common approach to astrophysical GW inference is to use agnostic priors for some GW observables. For example, analyses of the first two observed BNS mergers GW170817 (e.g., Abbott et al. 2017, 2018, 2019) and GW190425 (e.g., Abbott et al. 2020) employed uniform priors on the chirp mass $\mathcal{M}$ and the dimensionless tidal deformability $\tilde{\Lambda}$, implicitly assuming these parameters are uncorrelated. These are the two most informative parameters for the neutron star EOS.

The above prior assumptions are overly conservative. Basic physical principles such as causality, consistency with nuclear-physics experiments, and perturbative QCD calculations, imply a strong correlation between $\mathcal{M}$ and $\tilde{\Lambda}$ in neutron-star binaries. Constraints from astrophysical measurements of neutron-star masses and radii further tighten this correlation (Altiparmak et al. 2022; Ecker & Rezzolla 2022; Ecker & Rezzolla 2022).

Motivated by this, we introduce physics-informed joint priors on $\mathcal{M}$ and $\tilde{\Lambda}$ that incorporate microphysical constraints from nuclear theory and perturbative QCD, as well as astrophysical constraints on neutron-star masses and radii into astrophysical GW parameter inference. We show a significant improvement on the measurement of the marginalised posterior on $\tilde{\Lambda}$ from the GW observation of GW170817 from $\tilde{\Lambda}_{1.186} < 720$ with the uniform priors to $\tilde{\Lambda}_{1.186} = 384^{+306}_{-158}$ with the new physics-informed prior (confidence intervals are 90% throughout).

We infer a posterior distribution on EOS parameters and find that the neutron-star radius measurement for a $1.4\,M_\odot$ star improves from $R_{1.4} = 12.54^{+1.05}_{-1.54}\,\mathrm{km}$, as reported in the original LVK analysis (Abbott et al. 2019), to $R_{1.4} = 12.11^{+0.91}_{-1.11}\,\mathrm{km}$ when using physics-informed priors and a spectral EOS (see also Most et al. 2018; Capano et al. 2020, for some of the first estimates made after the GW170817 event). Similarly, for GW190425, the effective tidal deformability improves from $\tilde{\Lambda}_{1.44} = 381^{+1065}_{-271}$ to $\tilde{\Lambda}_{1.44} = 183^{+180}_{-100}$.

In addition to astrophysical inference, we show that these priors can be used to perform model selection, and hence distinguish between GW-source types. We perform model selection between a BNS and a neutron star – black hole (NSBH) origin for GW190425 and find a Bayes factor of 1.33 in favour of a BNS origin. When including observed rates as the prior odds, we find an odds ratio of 0.33, when considering the full population of NSBH mergers, indicating marginal support for a NSBH origin. When considering only mass-gap NSBH mergers, the odds ratio is 1.33, marginally in favour of a BNS merger origin. Regardless, we find that model selection for the origin of GW190425 remains uninformative.

In advocating for this prior becoming standard in the literature, we provide an open-source configuration for the `Bilby` Bayesian inference library (Ashton et al. 2019; Romero-Shaw et al. 2020) currently used for the majority of GW parameter estimation by the LIGO-Virgo-KAGRA (LVK) collaboration (e.g., Aasi et al. 2015; Acernese et al. 2015; Akutsu et al. 2020). This code provides a numerical implementation of the joint probability distribution as a `Bilby` constrained-prior object for $\mathcal{M}$ and $\tilde{\Lambda}$. We describe details of the method and implementation in the next Section.

2. METHODS

The construction of our physics-informed prior closely follows that of Altiparmak et al. (2022), which we briefly review here (see Ecker & Rezzolla 2022; Ecker & Rezzolla 2022, for additional details). We begin by building a large set of EOSs by stitching together various components. At the lowest densities ($n < 0.5\,n_s$) we adopt the Baym-Pethick-Sutherland (BPS) model (Baym et al. 1971) for the neutron-star crust. In the range $0.5\,n_s \leq n < 1.1\,n_s$, we randomly sample polytropes to span the range between the softest and stiffest EOSs from Hebeler et al. (2013). At high densities ($n \gtrsim 40\,n_s$), corresponding to a baryon chemical potential of $\mu = 2.6\,\mathrm{GeV}$, we impose the perturbative QCD constraint from Fraga et al. (2014) on the pressure $p(X, \mu)$ of cold quark matter, with the renormalization scale parameter X sampled uniformly in the range $[1, 4]$.

For the intermediate density range ($1.1\,n_s < n \lesssim 40\,n_s$), we use the parametrization method of Annala et al. (2020), which models the sound speed as a function of the chemical potential, $c_s^2(\mu)$, using piecewise-linear segments:

$$c_s^2(\mu) = \frac{(\mu_{i+1} - \mu)\, c_{s,i}^2 + (\mu - \mu_i)\, c_{s,i+1}^2}{\mu_{i+1} - \mu_i}\,, \qquad (1)$$

where μ_i and $c_{s,i}^2$ are parameters defining the i-th segment in the range $\mu_i \leq \mu \leq \mu_{i+1}$. In this section, we adopt natural units where $c = G = 1$, unless otherwise stated.

The number density follows as

$$n(\mu) = n_1 \exp\left(\int_{\mu_1}^{\mu} \frac{d\mu'}{\mu' c_s^2(\mu')}\right)\,, \qquad (2)$$

where $n_1 = 1.1\,n_s$, and $\mu_1 = \mu(n_1)$ is set by the corresponding polytropic EOS. The pressure is then obtained via

$$p(\mu) = p_1 + \int_{\mu_1}^{\mu} d\mu'\, n(\mu')\,, \qquad (3)$$

where the integration constant p_1 matches the pressure of the polytrope at $n = n_1$. We numerically integrate Eq. (3) using seven segments for $c_s^2(\mu)$.

Using this framework, we generate approximately 10^6 EOSs by randomly selecting the maximum sound speed $c_{s,\mathrm{max}}^2 \in [0, 1]$ and uniformly sampling the free parameters $\mu_i \in [\mu_1, \mu_{N+1}]$ (where $\mu_{N+1} = 2.6\,\mathrm{GeV}$) and $c_{s,i}^2 \in [0, c_{s,\mathrm{max}}^2]$. These EOSs are, by construction, consistent with nuclear theory and perturbative QCD uncertainties.

To incorporate astrophysical constraints, we solve the Tolman-Oppenheimer-Volkoff (TOV) equations for each EOS and retain only those satisfying the mass measurements of J0348+0432 (Antoniadis et al. 2013, $M = 2.01 \pm 0.04\,M_\odot$) and J0740+6620 (Cromartie et al. 2020; Fonseca et al. 2021, $M = 2.08 \pm 0.07\,M_\odot$), discarding EOSs with a maximum mass $M_{\mathrm{TOV}} < 2.0\,M_\odot$. In addition, we impose NICER radius constraints from J0740+6620 (Miller et al. 2021; Riley et al. 2021) and J0030+0451 (Riley et al. 2019; Miller et al. 2019), rejecting EOSs with $R < 10.75\,\mathrm{km}$ at $M = 2.0\,M_\odot$ or $R < 10.8\,\mathrm{km}$ at $M = 1.1\,M_\odot$. We note that, unlike what was done in Altiparmak et al. (2022) and Ecker & Rezzolla (2022), we do not impose any GW-informed constraints on the binary tidal deformability in the construction of our prior as we are reanalysing these data using the new prior.

In the BNS case, the binary tidal deformability of the prior is computed as

$$\tilde{\Lambda}_{\mathrm{BNS}} := \frac{16}{13}\frac{(12M_2 + M_1)M_1^4\Lambda_1 + (12M_1 + M_2)M_2^4\Lambda_2}{(M_1 + M_2)^5}\,, \qquad (4)$$

where M_i, R_i, and $\Lambda_i := \frac{2}{3}k_2\,(R_i/M_i)^5$ are the component source-frame masses, radii, and tidal deformabilities, respectively, and k_2 is the second tidal Love number. The source-frame chirp mass is $\mathcal{M} := (M_1 M_2)^{3/5}(M_1 + M_2)^{-1/5}$ and we define the mass ratio $q := M_2/M_1$ with $M_2 \leq M_1$.

The tidal deformability of a black hole is zero $\Lambda_{\rm BH} = 0$, implying the total tidal deformability for an NSBH binary is

$$\tilde{\Lambda}_{\rm NSBH} := \frac{16}{13}\frac{(12M_1 + M_2)M_2^4\Lambda_2}{(M_2 + M_1)^5}\,, \tag{5}$$

where M_2 and Λ_2 are the neutron-star mass and tidal deformability, respectively, and M_1 is the black-hole mass.

2.1. *BNS and NSBH mergers*

We use the probability distributions described in the previous Section as priors on the source-frame chirp mass and effective tidal deformability. These priors are shown in Fig. 1 where the top oanel is the BNS prior and the bottom panel is the NSBH prior. We implement this in `Bilby` using the `conditional_prior` function, interpolating a 200×200 grid to draw from a joint prior on $\mathcal{M}$ and $\tilde{\Lambda}$. The 90% probability contours (dashed black curves) can be fit by a simple power law

$$\tilde{\Lambda}_{\rm min(max)} = a + b\,\mathcal{M}^c\,, \tag{6}$$

where the fitting coefficients are $a = -57\,(-23)$, $b = 583\,(2690)$ and $c = -3.9\,(-4.5)$ for the BNS case and $a = 6\,(-5)$, $b = 296\,(1700)$ and $c = -6.6\,(-4.9)$ for the NSBH case. For comparison we also show contours for $a = -41\,(-0.55)$, $b = 539\,(2091)$ and $c = -4.5\,(-5.5)$ (dashed grey curves) corresponding to a prior in which the constraint on the tidal deformability from the GW170817 event, i.e., $\tilde{\Lambda} < 720$, is imposed; this corresponds to the analysis in Altiparmak et al. (2022) and highlights that while the estimate of $\tilde{\Lambda}_{\rm min}$ is hardly affected, that for $\tilde{\Lambda}_{\rm max}$ is significantly smaller when imposing the GW170817 constraint.

We sample directly in source-frame chirp mass, mass ratio and the two tidal deformability parameters $\tilde{\Lambda}$ and $\delta\Lambda$.[1] We note that q is not a variable that depends on the EOS, and we therefore take the standard approach of choosing a prior on q uniform in the range $q \in [0.125,\,1]$ (e.g., Romero-Shaw et al. 2020). The choice of a lower limit of $q = 0.125$ is due to waveform limitations and is not astrophysically motivated.

While both $\tilde{\Lambda}$ and $\delta\tilde{\Lambda}$ are related to the EOS, we only impose a prior here on $\tilde{\Lambda}$, as $\tilde{\Lambda}$ is measured much better than $\delta\tilde{\Lambda}$ (e.g., Wade et al. 2014). In principle, one could set a combined prior on the three parameters $(\mathcal{M},\,\tilde{\Lambda},\,\delta\tilde{\Lambda})$, however, adding this latter parameter would not add significantly to parameter estimation in the relatively low signal-to-noise regime. We therefore leave exploration of physically-motivated priors on $\delta\tilde{\Lambda}$ for future work.

The above discussion only refers to BNS mergers as the corresponding parameterisation does not go to large enough

[1] The quantity $\delta\tilde{\Lambda}$ is another tidal deformability parameter that enters at sixth-order post-Newtonian, cf. $\tilde{\Lambda}$ that enters at fifth order (see Favata 2014; Wade et al. 2014, for definitions of the these quantities).

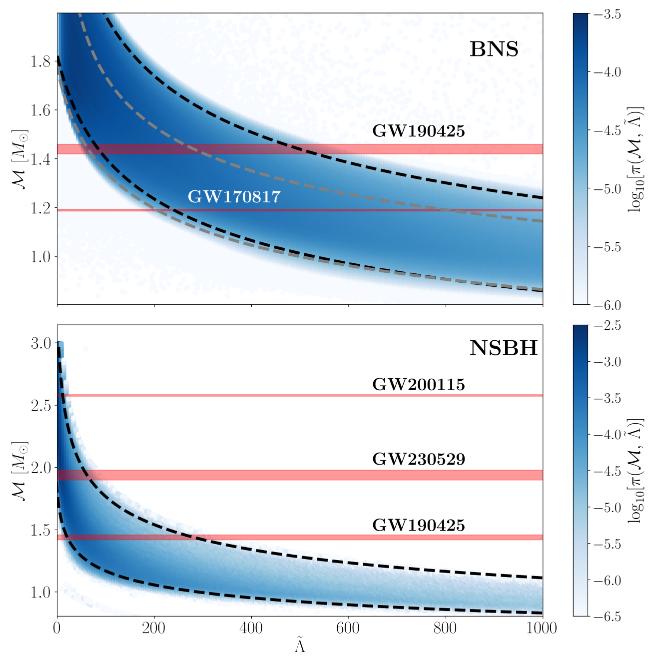

Figure 1. Priors for the source-frame chirp mass $\mathcal{M}$ and the dimensionless tidal deformability $\tilde{\Lambda}$ in the case of BNS (top panel) or NSBH binaries (bottom panel). In both panels, dashed black curves indicate analytic fits to the 90% contours of the distribution given in Eq. (6), while the dashed grey curves are the corresponding fits obtained when imposing the GW170817 constraint on the prior following Altiparmak et al. (2022). Horizontal red bands represent the 90% credible intervals for the measured chirp masses of GW170817 (Abbott et al. 2017), GW190425 (Abbott et al. 2020), GW200115 (Abbott et al. 2021), and GW230529 (Abac et al. 2024).

chirp-mass values for NSBHs. Hence, in the case of NSBH mergers we derive a new distribution by going back to the raw distributions of progenitor stellar masses and radii, and combine that progenitor distribution with that of a BH of mass M_1 and tidal deformability $\Lambda_1 = 0$. In principle, in this case we could use more sophisticated and realistic mass-ratio distributions, such as those informed by state-of-the-art binary population synthesis (e.g., Broekgaarden et al. 2021). In practice, we use a uniformly-distributed mass-ratio prior as we find that our results do not depend significantly on the choice made for the mass-ratio prior distribution. Finally, also for NSBHs we choose to work with priors in $\mathcal{M}-\tilde{\Lambda}$, enforcing $\tilde{\Lambda}(\Lambda_2, \mathcal{M}, q)$ where Λ_2 is the tidal deformability of the neutron star.

2.2. *Parameter estimation*

We perform Bayesian parameter estimation on the GW strain data of the two BNS merger events GW170817 and GW190425 using the publicly-available data from the

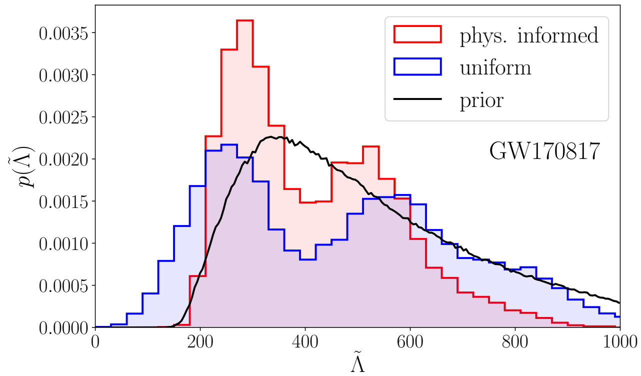

Figure 2. Marginalised posterior distributions of the tidal deformability $\tilde{\Lambda}$ for GW170817. The posterior using a physics-informed prior is shown in red, while the posterior using uniform priors from the LVK analysis is shown in blue: we show the (marginalised) prior with the black curve. Note that the posterior from the LVK shows support up to $\tilde{\Lambda} = 2000$.

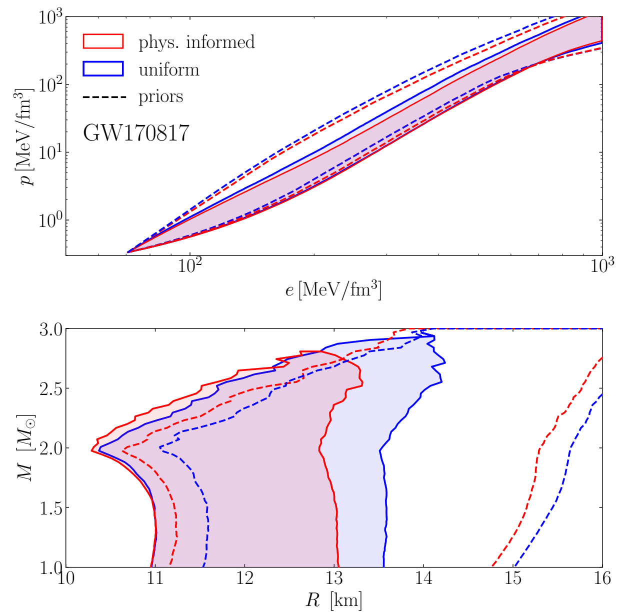

Figure 3. Posterior distributions of the range of pressures and energy densities for a nuclear-matter EOS (top panel) and of the corresponding masses and radii (bottom panel) in the case of GW170817. In both panels, the shaded regions show the 90% credible intervals of the posterior distributions obtained using the physics-informed priors (red) or the uniform priors (blue), while the dashed curves show the 90% credible intervals of the priors on chirp mass and tidal deformability when mapped to the spectral parameterisation.

GWOSC repository[2]. We utilise the `Bilby` Bayesian inference framework (Ashton et al. 2019; Romero-Shaw et al. 2020) and priors on source–frame chirp mass and tidal–deformability as discussed in Section 2. Priors for all remaining parameters assume the so-called low-spin ($\chi \leq 0.05$) default priors as defined in `Bilby`. We use Planck 2018 cosmological parameters (Planck Collaboration et al. 2020) such that Hubble's constant and the matter density are respectively $H_0 = 67.66\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_M = 0.30966$.

Parameter estimation for each run is performed using the `dynesty` nested sampler (Speagle 2020) and the `IMRPhenomPv2_NRTidal` waveform (Dietrich et al. 2019). Analysis of BNS merger signals is computationally expensive and as such we utilise parallel implementation of `Bilby`, i.e., `pBilby` (Smith et al. 2020). We reconstruct pressure–density and mass–radius curves from posterior samples of source–frame chirp mass, mass ratio, and effective tidal deformability by performing EOS inference on the posterior samples of mass and tidal deformability. We utilise a four-piece spectral parameterisation (Lindblom 2010) stitched to an SLy (Douchin & Haensel 2001) crust at low densities (further details can be found in Appendix A).

The choice of adopting an approach for the EOS inference (i.e., four-piece spectral parameterisation) that is different from that used in the EOS construction (i.e., seven-segment sound-speed parameterisation) follows a precise logic. In particular, this has the advantage that it is possible to compare directly with the LVK inference analysis and hence clearly distinguish the assumptions behind the definition of the priors from those employed in the inference. At the same time, this also implies that the posteriors obtained for the EOSs parameters differ from the corresponding priors. While these differences are small, we plan to explore the use of the same set of assumptions for the EOS priors and inference in future work.

[2] https://gwosc.org

3. RESULTS

Figure 2 shows the posterior distributions of the tidal deformability of GW170817 analyzed using the physics-informed BNS (red), the publicly-released uniform priors from the LVK collaboration (blue; Abbott et al. 2017), and the marginalised prior for $\tilde{\Lambda}$ (black). While we show only the tidal posteriors, we sample over all binary parameters. Posterior distributions for all other binary parameters, including the chirp mass and mass ratio, are identical for the physics-informed and uniform-prior runs. Note that, as expected, the marginalised posterior is not significantly different at large tidal deformabilities (compare blue and black curves), but shows significant changes at small values of $\tilde{\Lambda}$, essentially excluding values $\tilde{\Lambda} \lesssim 200$ for the physics informed priors. As a result, the 90% lower and upper limits on the tidal deformability are now $226 \leq \tilde{\Lambda}_{1.186} \leq 690$. These arguably represent the most interesting consequence of the physics-informed prior on the $\mathcal{M}-\tilde{\Lambda}$ relation.

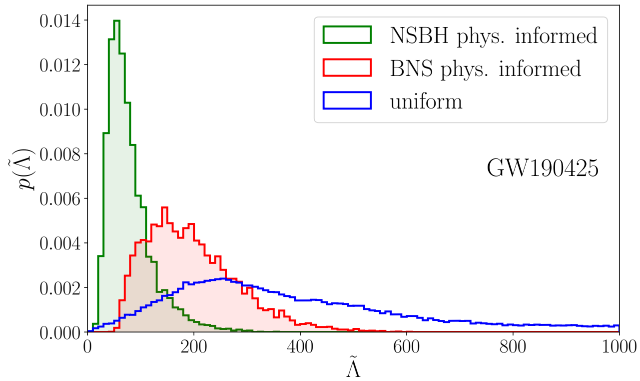

Figure 4. The same as in Fig. 2 but for the GW190425 event. The posterior using a physics-informed prior for BNS mergers is shown in red, while the posterior for NSBH mergers is shown in green. The posterior using uniform priors from the LVK is reported in blue.

Figure 3 shows the impact of the $\mathcal{M} - \tilde{\Lambda}$ relation on the reconstructed posterior distributions of the pressure and energy density (top panel) and of the mass and radius (bottom panel) relative to the GW170817 event. More specifically, the shaded regions show the 90% credible intervals for the posteriors samples obtained using the physics-informed priors (red) and uniform priors (blue). In this way, we can improve the constraints on the radius of a $1.4\,M_\odot$ neutron star from the values provided by LVK $R_{1.4} = 12.54^{+1.05}_{-1.54}\,\mathrm{km}$ to $R_{1.4} = 12.11^{+0.91}_{-1.11}\,\mathrm{km}$. The dashed curves in Fig. 3 represent the 90% credible intervals of the priors on the physics-informed prior (red) and on the uniform prior (blue) when sampled with the spectral parameterisation.

Figure 4 offers a complementary view to Fig. 2 and shows the posterior distribution of the intrinsic tidal parameter of GW190425 when analyzed using both the BNS prior (red) and NSBH prior (green; see Fig. 1). We show the posterior distribution of the LVK collaboration obtained using uniform priors in blue (Abbott et al. 2020).

Bayesian-model selection between the BNS and NSBH merger for the GW190425 event gives a Bayes factor of 1.33. We calculate the odds ratio, the product of the Bayes factor and our prior odds ratio. The latter can be calculated as a ratio of observed events, in our case the ratio of observed BNS and observed NSBH mergers. The total number of BNS mergers (excluding GW190425) is one; and the total number of publicly-released confirmed NSBH mergers is three: GW200105, GW200115 (Abbott et al. 2021), GW230529 (Abac et al. 2024). Thus the prior odds ratio is $1/3$ and the odds ratio is 0.44, which represents marginal evidence for a NSBH merger. To appreciate how marginal this estimate is, we might argue that only mass-gap events like GW230529 should be included in the prior. In this case, the prior odds is unity and the odds ratio is again 1.33, in favour of a BNS origin. Regardless, any choice of prior odds remains uninformative for determining the nature of GW190425 and additional observables [e.g., on the properties of the ejected matter (Most et al. 2021)] need to be included in the priors.

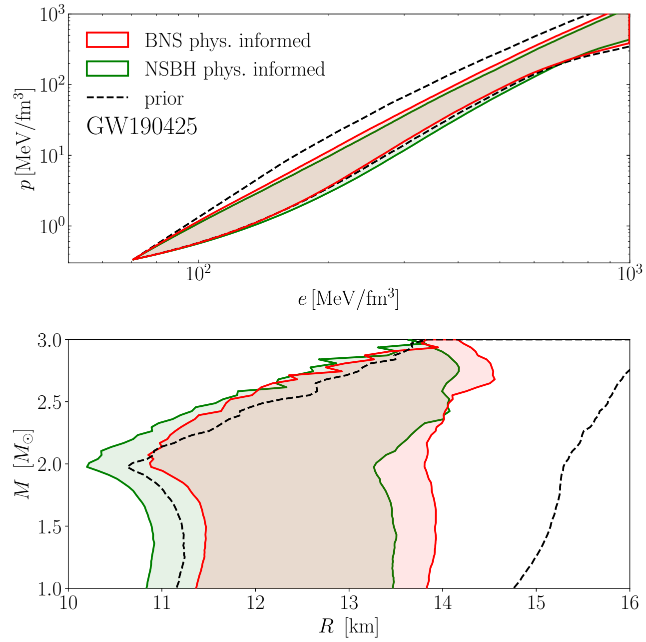

Figure 5. The same as in Fig. 3 but for the GW190425 event. The top panel shows the posterior distributions in pressure–density space, while the bottom panel shows the reconstructed mass–radius curves. The shaded regions show the 90% credible intervals of the posterior distributions obtained from the analysis using the physics-informed BNS prior (red) and the NSBH prior (green). The black dashed curve shows the 90% credible intervals of the priors when mapped to a spectral parameterisation.

Figure 5 is logically equivalent to Fig. 3, but contrasting BNS and NSBH merger interpretations for the GW190425 event. Hence, it shows the reconstructed pressure–density (top) and mass–radius (bottom) posteriors of GW190425. The EOS 90% credible intervals of posteriors obtained using the BNS physics-informed prior are shown in red, while the NSBH posteriors are shown in green. Once again, the dashed curve represents our prior on the EOS. Overall, we find that the posterior distributions constrain $R_{1.4} = 12.69^{+1.24}_{-1.22}\,\mathrm{km}$ using the BNS prior and $R_{1.4} = 12.06^{+1.44}_{-1.15}\,\mathrm{km}$ using the NSBH prior.

4. CONCLUSIONS

The still poor knowledge of the EOS of nuclear matter represents a significant obstacle in the analysis of GW events involving neutron stars. This is true not only for parameter estimation of neutron stars in the case of BNS mergers, but also to assess whether the low-mass component is actually a neutron star or a black hole in candidate NSBH mergers. To counter these limitations, a number of approaches have recently been developed that use a large number of physically and astrophysically consistent parameterised EOSs to deduce quasi-universal behaviours of the neutron star. A particularly useful relation is the one presented by Altiparmak et al. (2022), who have shown a tight statistical correlation between the chirp mass $\mathcal{M}$ – arguably, the best-measured quantity in binary mergers – and the binary tidal deformability $\tilde{\Lambda}$.

We here exploit the $\mathcal{M}-\tilde{\Lambda}$ relation to re-investigate the constraints on the GW170817 and GW190425 events. More specifically, we employ the reference LVK pipeline analysis where uniform priors are replaced by the physics-informed priors coming from the EOS parameterisation. In this way, we find that a significant improvement in the inferred properties is obtained in the case of GW170817, where the constraints on the radius of a $1.4\,M_{\odot}$ neutron star improve from $R_{1.4} = 12.54^{+1.05}_{-1.54}\,\mathrm{km}$ to $R_{1.4} = 12.11^{+0.91}_{-1.11}\,\mathrm{km}$. Similarly, the constraints on the tidal deformability are considerably restricted, going from $\tilde{\Lambda}_{1.186} < 720$ to $\tilde{\Lambda}_{1.186} = 384^{+306}_{-158}$. In the case of GW190425, we assess the nature of the secondary in the binary by computing the odds ratio associated to a neutron star or a black hole hypothesis. We find this analysis to be uninformative, with a odds ratio between 0.44 and 1.33 in favour of the BNS merger hypothesis. In either case, we improve the constraints on the radius estimates with respect to previous LVK with the posterior distributions indicating $R_{1.4} = 12.69^{+1.24}_{-1.22}\,\mathrm{km}$ for BNS and $R_{1.4} = 12.06^{+1.44}_{-1.15}\,\mathrm{km}$ for NSBH mergers.

Overall, our results clearly indicate that physics-informed priors do improve the parameter estimation of GW events involving neutron stars and we thus advocate their use as versatile tools for which we provide an open-source code for reproducibility and further adaptation of the method. At the same time, what we present here can be improved in a number of ways. First, by adopting for the EOS inference the same EOS parameterisation employed in the construction of the physics-informed priors. Second, while we have constructed priors using binaries containing neutron stars with "low spins" (i.e., $\chi := J/M^2 \leq 0.05$, where J and M are the star's angular momentum and mass, respectively), the inference can be extended by considering also "high-spin" ($\chi \leq 0.89$) priors. Finally, in the case of NSBH mergers, the posteriors can be further constrained by restricting the high-range of neutron-star masses that would lead to a prompt black-hole formation and hence prevent an electromagnetic counterpart (Bauswein et al. 2013; Koeppel et al. 2019; Tootle et al. 2021; Kashyap et al. 2022; Kölsch et al. 2022; Ecker et al. 2025). We plan to explore these extensions in future work.

APPENDIX

A. EQUATION OF STATE INFERENCE

The posterior distirbution on the EOS parameters η from the GW data is

$$p(\epsilon|d) = \frac{p(d|\epsilon)\pi(\epsilon)}{\mathcal{Z}}\,, \tag{A1}$$

where $\pi(\epsilon)$ is the prior on the EOS parameters, and $\mathcal{Z}$ is the evidence. The likelihood is given by

$$p(d|\epsilon) = \int d\theta\, p(d|\theta)\,\pi(\theta|\epsilon)\,, \tag{A2}$$

where θ are our nuisance parameters which we marginalise over. Our likelihood reduces to

$$p(d|\epsilon) = \int d\theta\; p(d|\mathcal{M},q,\tilde{\Lambda})\;\pi(\mathcal{M},q,\tilde{\Lambda}|\epsilon,\theta)\;d\mathcal{M}\;dq\;d\tilde{\Lambda}\,. \tag{A3}$$

Since we a dealing with a small number of events (i.e., $1-2$), we may ignore the modelling of overall neutron-star population parameters without introducing significant biases into the EOS inference (e.g., Agathos et al. 2015; Wysocki et al. 2020; Landry et al. 2020). The marginalised likelihood transforms to

$$p(d|\epsilon) = \int\; p(d|\mathcal{M},q,\tilde{\Lambda})\;\pi(\mathcal{M},q,\tilde{\Lambda}|\epsilon)\;d\mathcal{M}\;dq\;d\tilde{\Lambda}\,. \tag{A4}$$

In expression (A4) there are two terms that need to be calculated. The first one is $\pi(\mathcal{M},q,\tilde{\Lambda}|\epsilon)$ that represents a prior on chirp mass, mass-ratio and $\tilde{\Lambda}$ for a given EOS. The second one is the likelihood $p(d|\mathcal{M},q,\tilde{\Lambda})$. Typically, this likelihood would be obtained by reweighing posterior samples of parameter estimation runs

$$p(d|\mathcal{M},q,\tilde{\Lambda}) = \frac{p(\mathcal{M},q,\tilde{\Lambda}|d)}{\pi_{\mathrm{PE}}(\mathcal{M},q,\tilde{\Lambda})}\,, \tag{A5}$$

however if we did this we would be removing any of our prior information from the posterior samples. Since our prior on $\tilde{\Lambda}$ and $\mathcal{M}$ is actually physically motivated, we set

$$\pi_{\rm PE}(\mathcal{M}, q, \tilde{\Lambda}) = \pi(\mathcal{M}, q, \tilde{\Lambda}|\epsilon)\,, \tag{A6}$$

which then implies that

$$p(d|\epsilon) = \int p(\mathcal{M}, q, \tilde{\Lambda}|d)\; d\mathcal{M}\; dq\; d\tilde{\Lambda}\,. \tag{A7}$$

Since the chirp mass is often very-well constrained by parameter estimation, we take the median value of chirp mass $\bar{\mathcal{M}}$, and rewrite our posterior distribution as

$$p(\mathcal{M}, q, \tilde{\Lambda}|d) = p(q, \tilde{\Lambda})\; \delta(\mathcal{M} - \bar{\mathcal{M}})\,, \tag{A8}$$

which simplifies the integral to

$$p(d|\epsilon) = \int p(\bar{\mathcal{M}}, q, \tilde{\Lambda}|d)\; dq\; d\tilde{\Lambda}\,. \tag{A9}$$

We note that we can also express the tidal deformability as a function of $\bar{\mathcal{M}}$, q and ϵ, which removes another dimension from our integration, namely,

$$p(d|\epsilon) = \int p(q, \tilde{\Lambda}(\bar{\mathcal{M}}, q, \epsilon)|d)\; dq\,, \tag{A10}$$

which is now just an integral over the mass ratio q.

We evaluate the probability density of our marginalised posterior samples using a bounded kernel-density estimate and parameterise our EOS using the spectral parameterisation of Lindblom (2010). We use uniform priors for each EOS parameter and enforce a minimum TOV mass $M_{\rm TOV} \geq 2.0 M_{\odot}$ and causality.

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DATA AVAILABILITY

The implementation of our physics-informed priors into the `Bilby` Bayesian inference library, along with documentation and example scripts can be found at https://github.com/spencermagnall/physics_informed_priors.

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